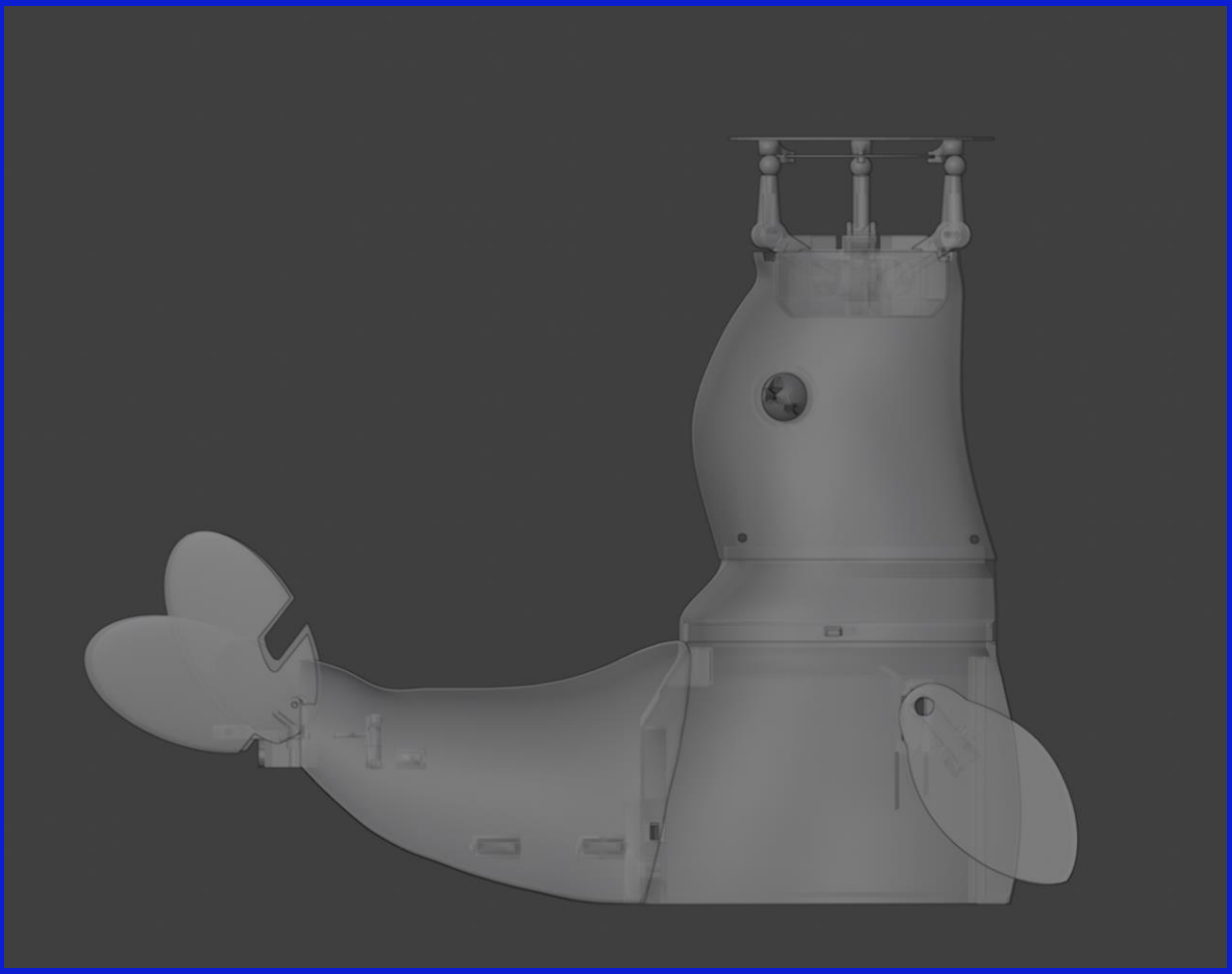
CYBER PHYSICAL SYSTEMS



PROJECT SEALIA

GROUP 19

ADAM SAINT- JAQUES, AMBER JIANG &

VALERIA AGUIRRE

GITHUB

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01

OVERALL CONCEPT

Sealia is an interactive self-balancing robotic seal that combines mechanical design, machine learning, movement, and sound to create a character-driven robotic experience. The project is based on the open-source BABOT platform, which has been adapted and expanded with a custom seal-inspired body, moving arms, an articulated tail, a ball platform, a launching mechanism, audio feedback, and an integrated vision system.

The main concept behind Sealia is to explore how a robot can respond to simple visual inputs with expressive physical behaviours. An external webcam positioned within the ball platform observes red, green, and blue balls. RGB values from the camera are used as inputs for a machine-learning model in Wekinator, which classifies the detected ball into one of three classes. A Processing script then sends the detected class to an Arduino through serial communication. The Arduino uses this information to trigger a unique performance made up of coordinated arm movements, tail movements, platform actions, ball launching, and sound.

Each colour gives Sealia a different personality and response. The red ball produces a happy and energetic performance; the green ball triggers a slower sequence that ends with the ball being “dropped” from the platform, and the blue ball creates an alternating arm-wave performance. By combining machine learning with mechanical expression, Sealia transforms a technical robotic system into an interactive character whose behaviour changes depending on what it sees.



3D Render of Sealia Concept. First Iteration.

02

ROBOT ARM MODIFICATIONS

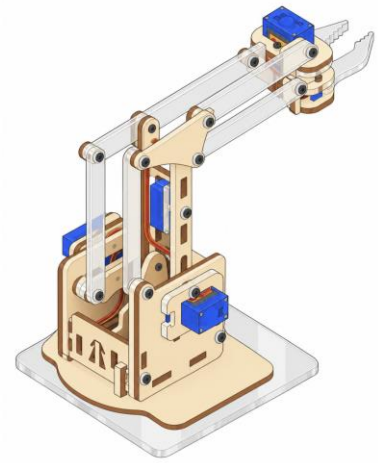
The original robot arm provided inspiration for how Sealia's movements could be created using servo-driven mechanisms. Rather than reproducing the arm directly, we adapted the principles of its joints, linkages, and geared motion to create movement suited to the seal character.

Sealia uses six servos in total. Three servos control the three arms of the balancing platform, allowing it to move between different programmed positions. The other three servos control the seal's two fins and tail.

The fin movement was inspired by the joint motion of the original robot arm. Servo rotation is transferred through the mechanism to move the fins up and down, creating expressive waving movements. The tail mechanism was inspired by the opening and closing action of the robot arm's gripper. This was adapted into a clapping motion, allowing the two sides of the tail to open and close during Sealia's performances.

This modification allowed us to take the mechanical principles demonstrated by the original robot arm and apply them to a completely different character and interaction.

Shown below, our original assessment of robot arm and inspiration for Selia.





03

MECHANICAL DESIGN

A quick reference to the motor pins, tail and flipper mechanisms, and the three-servo balancing platform

Motor pin reference Each motor is identified by a letter from A to F.

Motor	Function	Arduino pin
A	Seal flipper A	10
B	Seal flipper B	11
C	Seal tail	9
D	Balancing platform motor D	2
E	Balancing platform motor E	3
F	Balancing platform motor F	4

Mechanisms

Tail mechanism – paired gears create an opening and closing movement.

Seal Hand Flipper mechanism – a servo horn, bearing and bridge drive the attached flipper.

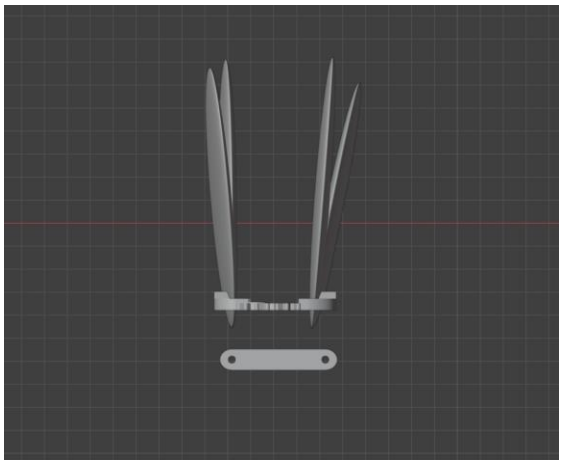
Balancing platform – three linkages and magnetic ball joints allow the acrylic platform to tilt.

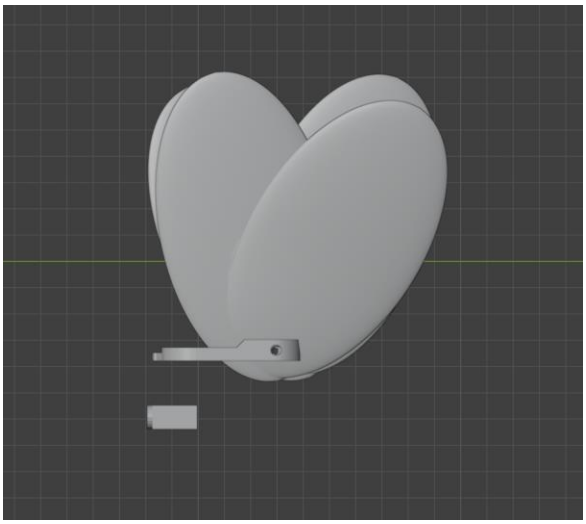
Tail Mechanism

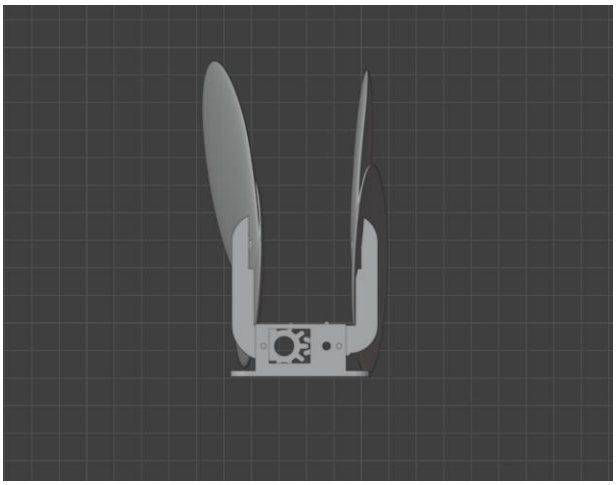
The seal tail adapts the opening and closing principle of a robot arm gripper.

A servo rotates one gear, which drives a second gear in the opposite direction. This moves the two sides of the tail together to open or close them. The flippers and tail were adapted from the robot arm gripper mechanism.









Tail Installation

The tail mechanism installed inside the seal body.

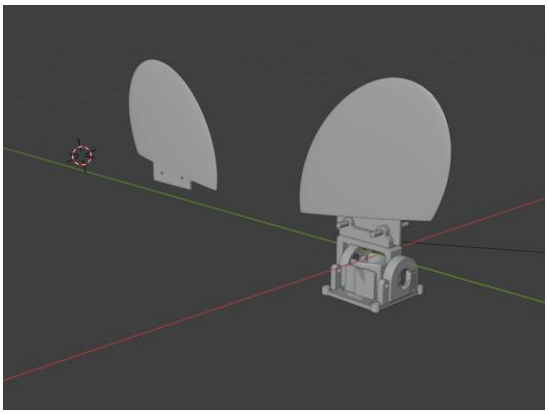


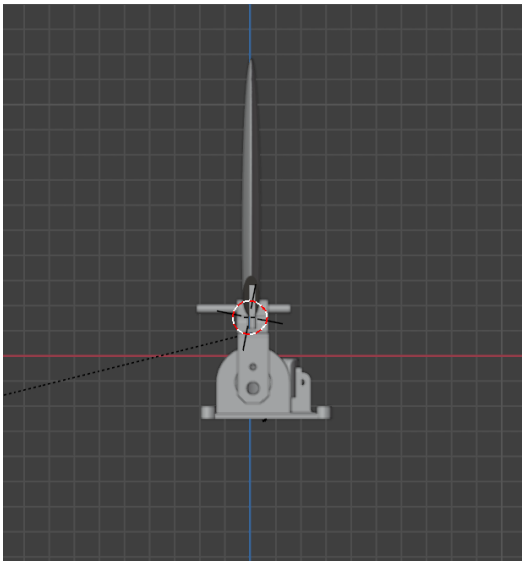
Tail assembly inside the seal, showing the servo and paired gears between the two tail sections.

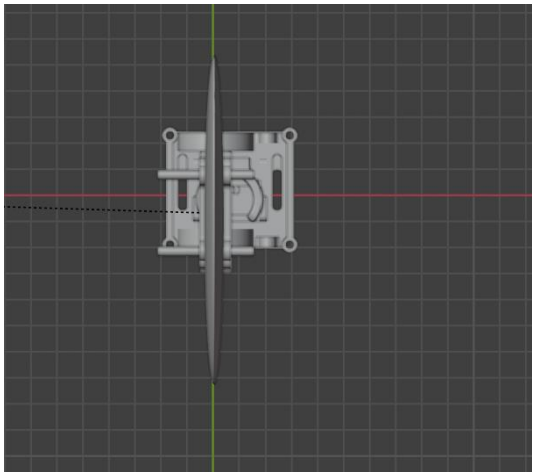
Seal Hand Flipper Mechanism

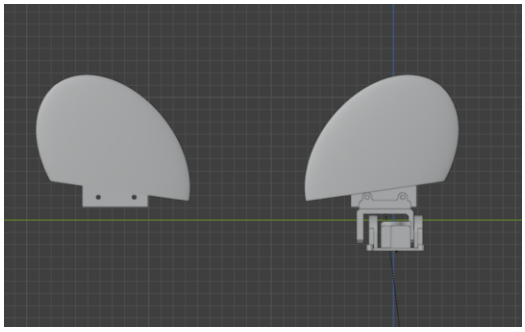
The flipper is a separate component bolted onto a servo-driven bridge.

The servo horn connects to one end of the mechanism. A bearing supports the opposite end and allows smooth rotation. A central bridge links both sides so they move together, tilting the flipper as the servo rotates.

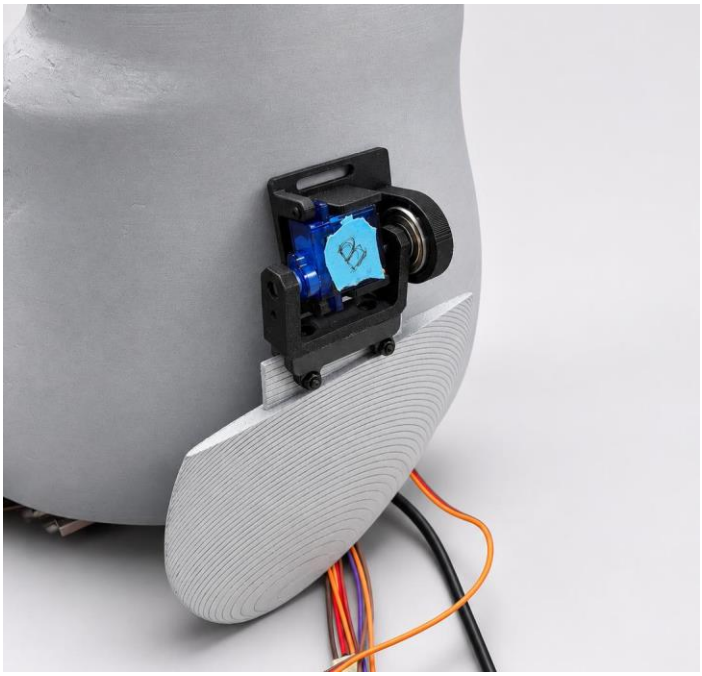


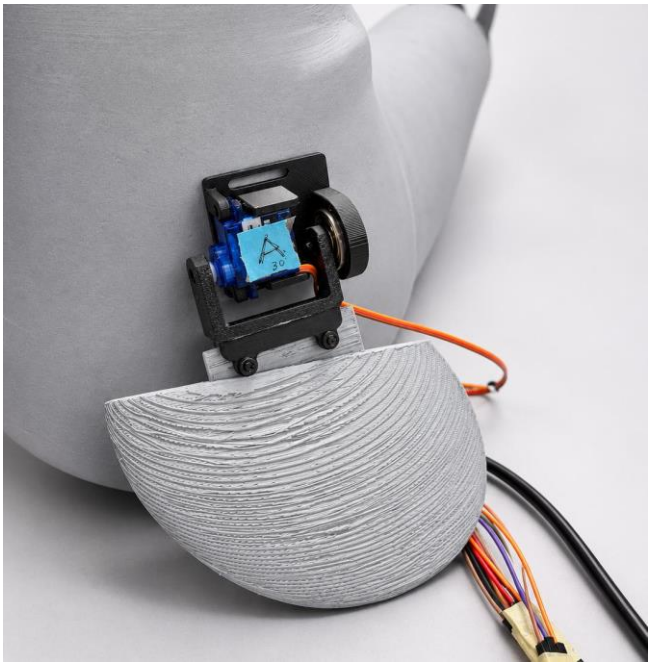






Flipper Installation





Servo and Output Pins

Component	Arduino Pin	Movement
Platform Servo D	Pin 2	0° up / 90° down
Platform Servo E	Pin 3	0° up / 90° down
Platform Servo F	Pin 4	0° up / 90° down
Tail Servo C	Pin 9	100° closed / 170° open
Left Arm A	Pin 10	45° down / 130° up
Right Arm B	Pin 11	45° down / 130° up
Buzzer	Pin 12	Audio output

The seal arms begin at approximately 80°. The Arduino gradually moves the arm servos one degree at a time toward their target positions to create smoother movements.

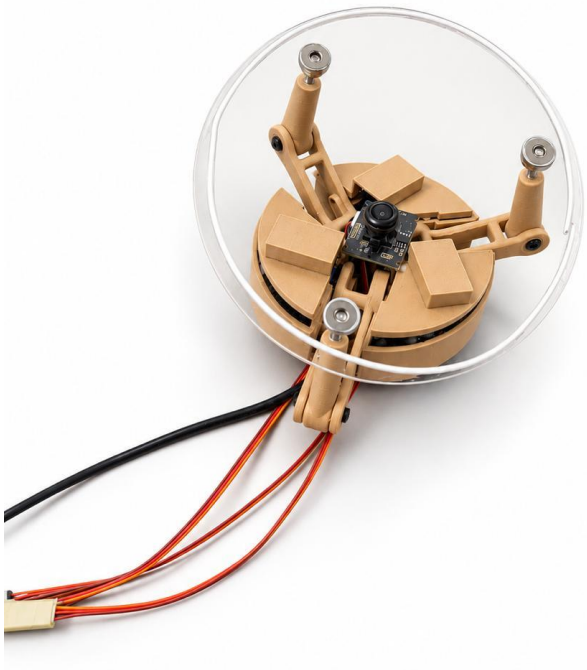
Balancing Platform

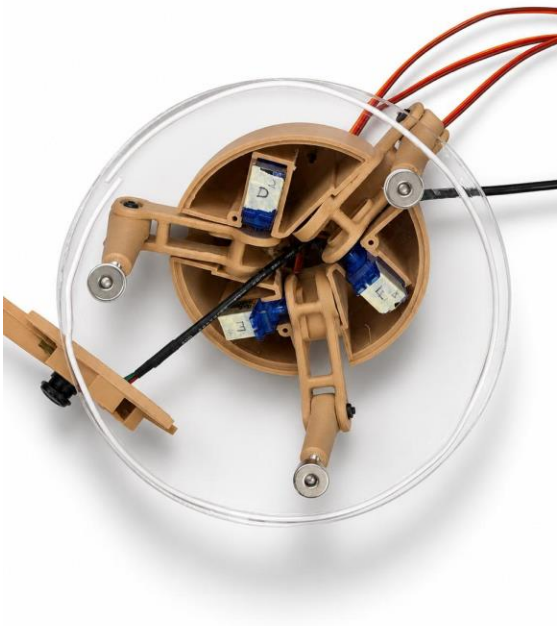
Three servo-driven linkages support and tilt the acrylic platform.

How it works

Each servo horn fits into a small mechanical arm, driving a linkage with pivoting joints. A ball at the top of each linkage connects magnetically to a countersunk magnet beneath the acrylic platform.

These magnetic ball joints allow the platform to tilt freely as the three servos adjust its position to balance the ball.

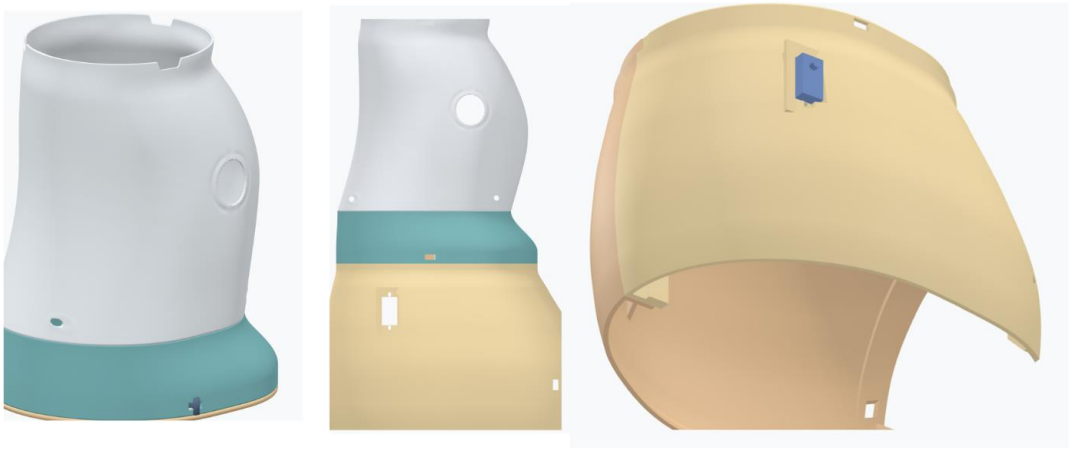


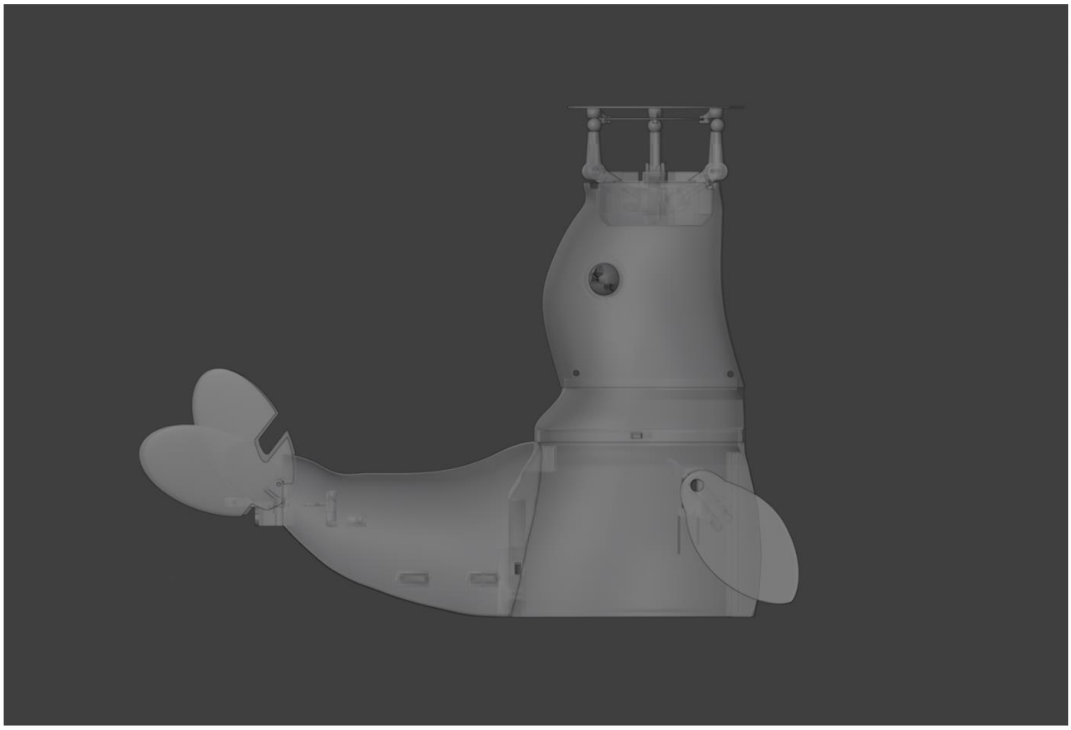


04

ROBOT SHELL

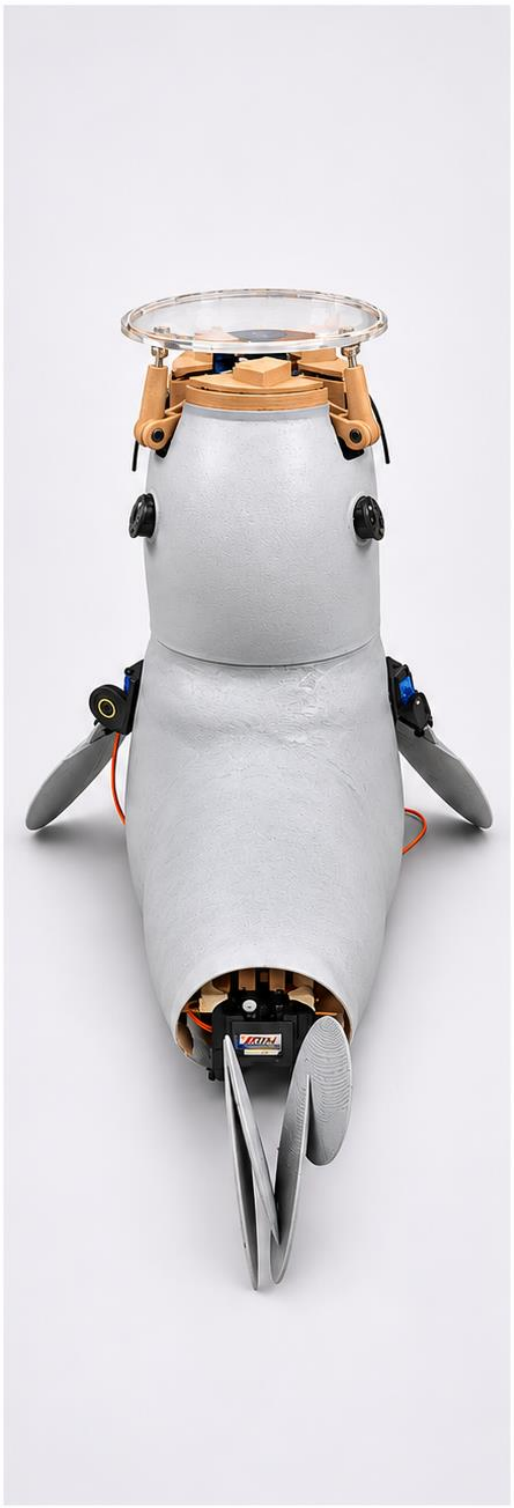
STL files for the head, neck, body and tail shell for Sealia.





FINAL DESIGN







05

SOFTWARE DESIGN

Sealia uses Wekinator, Processing, and Arduino together as one software pipeline. Original code files can be found in project Github.

```
External Webcam
    ↓
RGB Input
    ↓
Wekinator
    ↓
Class 1 / 2 / 3
    ↓
Processing
    ↓
Serial Communication
    ↓
Arduino
    ↓
Physical Performance
```

Wekinator

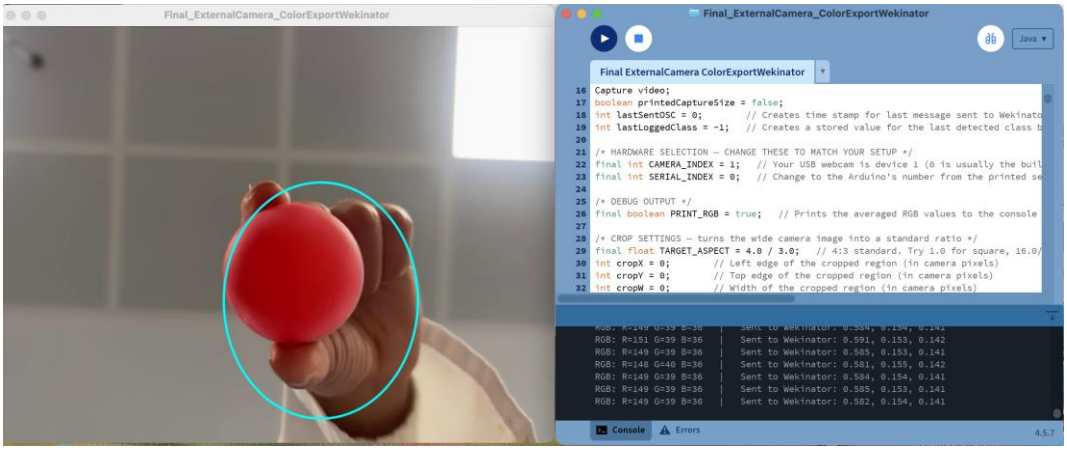
Wekinator is responsible for classifying the RGB input. It has been trained using examples from the red, green, and blue balls.

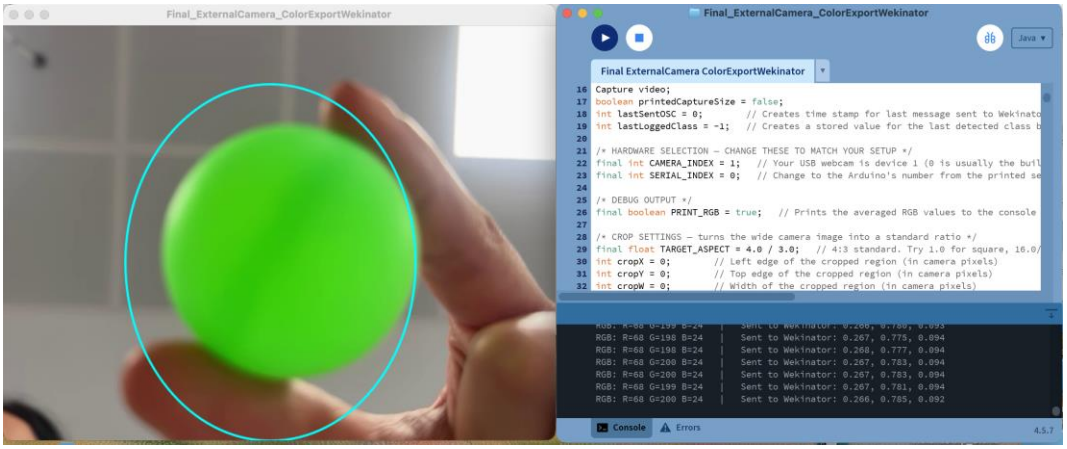
The output is one of three classes.

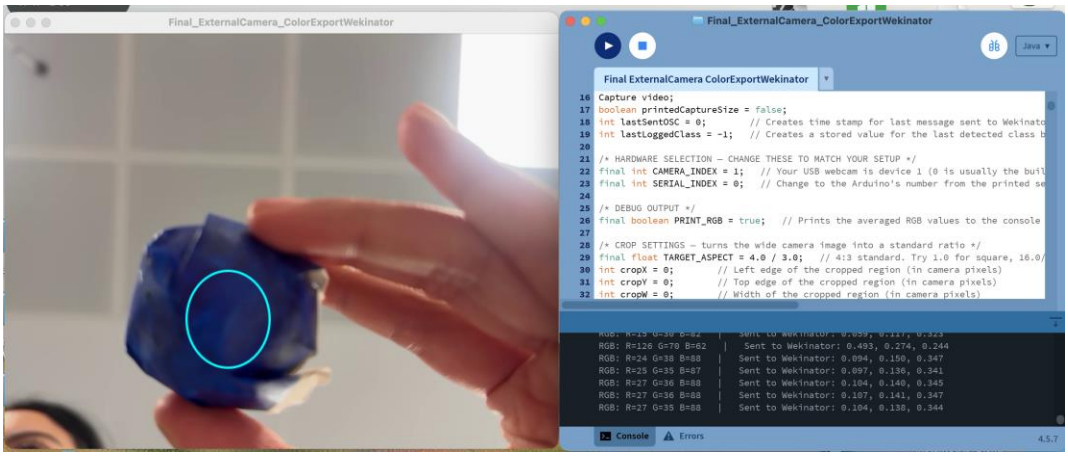
```
1 → Red (≤255, 0, 0)
2 → Green (0, ≤255, 0
3 → Blue (0, 0, ≤255)
```

Training Data

Wekinator was trained on camera input. The data was trained on video feedback of red, green and blue balls. Based on the highest value detected in a circular tracking, processing will determine the class.







Processing

Processing connects Wekinator to the Arduino.

The Processing script receives the class detected by Wekinator.

It then converts that class into a serial message and sends it to the Arduino.

```
Wekinator
  ↓
Detected Class
  ↓
Processing
  ↓
Serial Message
  ↓
Arduino
```

Arduino

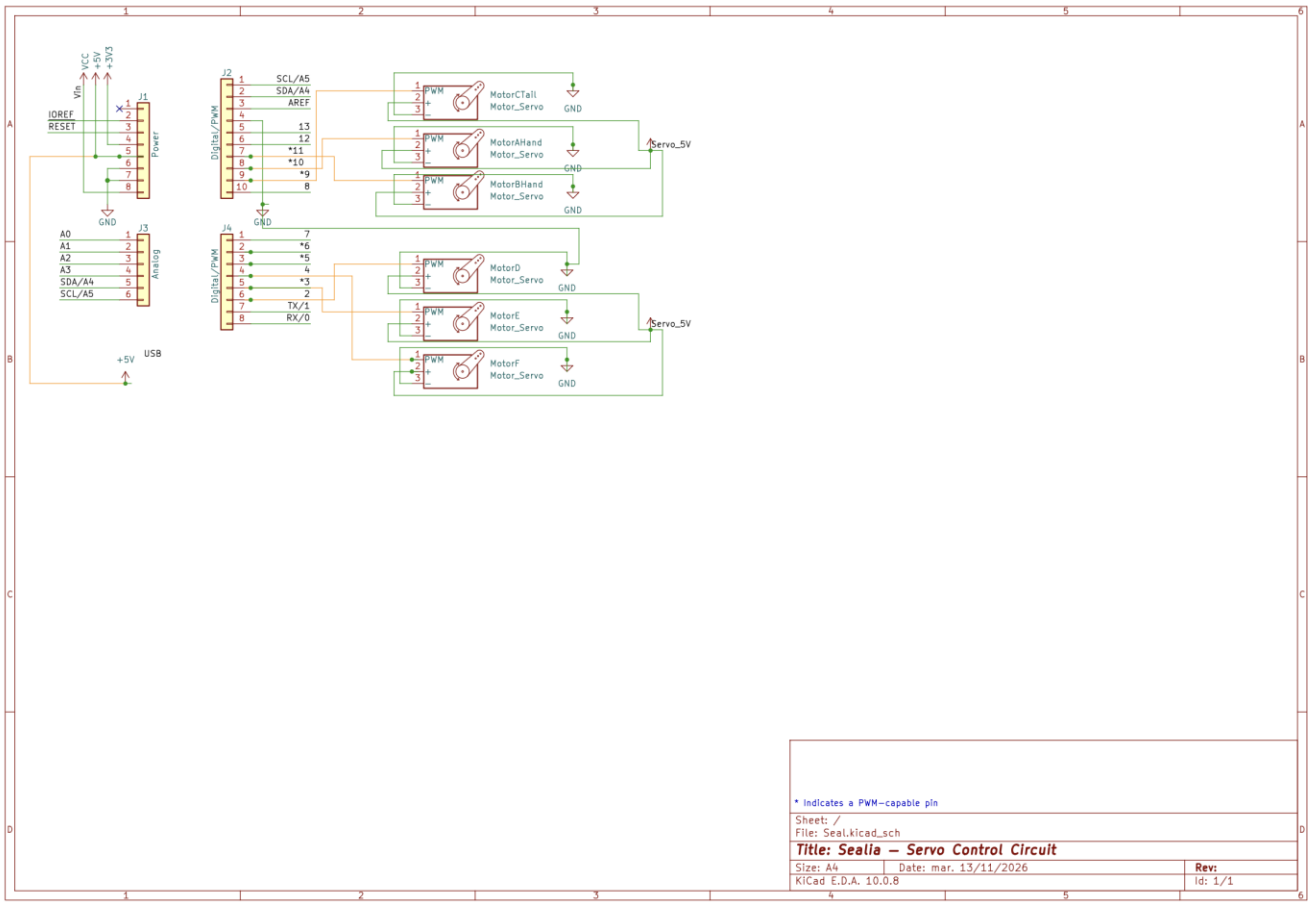
The Arduino receives the serial message and selects the corresponding performance.

It controls the physical outputs of Sealia, including:

- Arm movement
- Tail movement
- Ball platform position
- Audio playback
- Timing between movements

06

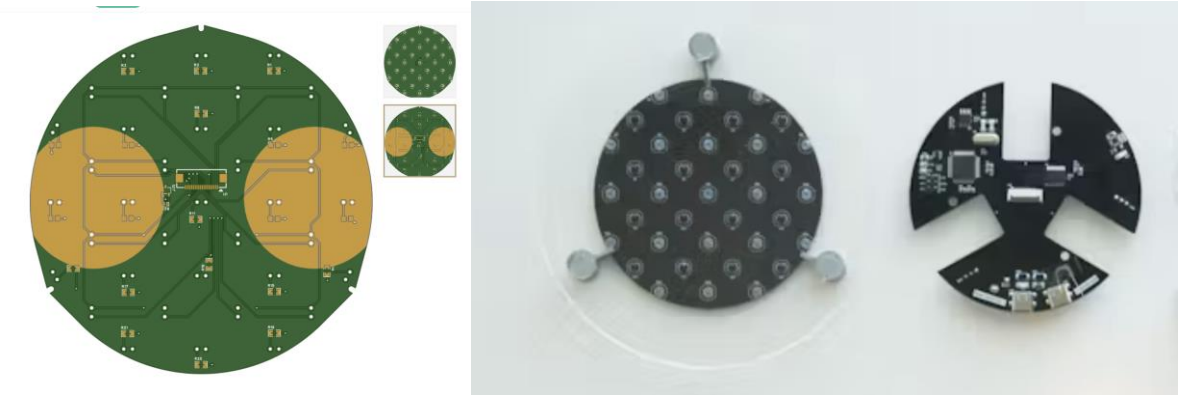
WIRING DIAGRAM



07

PROTOTYPE V1- HARDWARE CHANGES & ADJUSTMENTS

The first Sealia prototype was based on the self-balancing platform from the open-source BABOT project. BABOT uses a custom PCB and an infrared sensor array to detect the ball position and adjust the upper platform automatically.



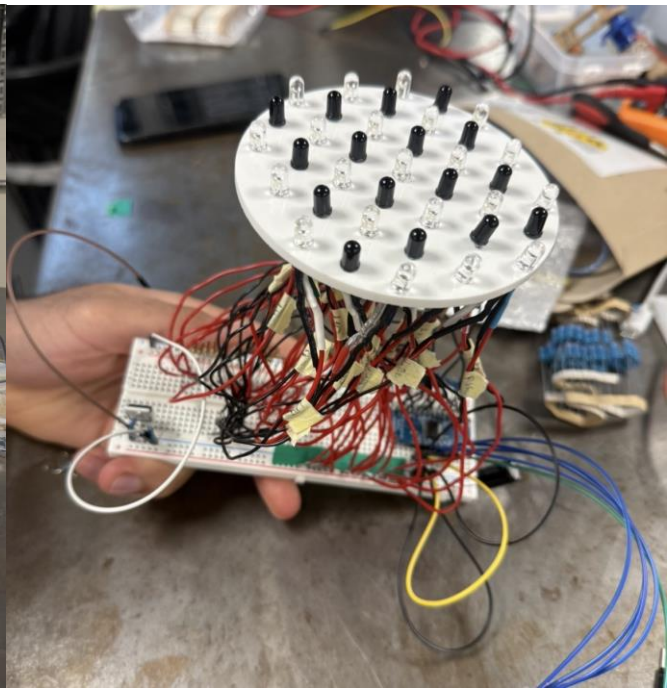
Pictured above, original custom PCB board and IR Sensor Board used by Babot.

We initially attempted to recreate this system by hand using separate IR LEDs and IR Emitters and receivers mounted into a laser-cut platform. A multiplexer and Arduino were planned to read the large number of sensor signals.

Although the sensor array was assembled, the amount of wiring, calibration, and additional hardware required made the system too complex to complete reliably within the available time and materials. The full IR balancing system was therefore moved to a future version of the project.

Pictured below, hand wired IR Sensor board.





Without automatic ball tracking, the three platform arms D, E, and F were instead controlled directly by the Arduino. The platform was programmed to perform three predefined motions: a flat lowered position, a launching motion, and a flat raised position.

This allowed the BABOT-inspired platform to remain part of Sealia while changing its purpose from automatic balancing to part of the robot's programmed performances. For the final v1, each servo was wired and powered by an external battery source.



08

AI STATEMENT

AI tools were used during the project as a support resource for troubleshooting and technical research. This included helping us understand how a custom PCB-based system could be recreated using individually wired components, as well as answering hardware-related questions during prototyping.

AI was also used to help explain electronic concepts, identify possible wiring approaches, and support problem-solving when issues occurred during development.

All final design decisions, hardware assembly, testing, programming, and implementation were completed by the project team.

09

FUTURE PROTOTYPE

Future development could include:

- A larger machine-learning training dataset
- Improved performance under different lighting conditions
- IR Sensor Board for actuated platform balancing

A larger training dataset containing examples from different lighting environments could make the colour classification more reliable in different lighting conditions. The current three-class system could also be extended so that Sealia responds to additional objects or interactions rather than only colour. The self balancing platform with further custom hardware will be completed and improved for a future version of this project.

10

AUTHORSHIP & CREDIT

Sealia is based on the open-source BABOT project created by JohanLink.

Original project:

https://github.com/JohanLink/BABOT

BABOT provides the foundation for the ball-balancing robot platform.

Sealia develops this platform further through:

- A custom seal-inspired body
- Modified mechanical components
- Moving arms
- An articulated tail
- A ball launching system
- Webcam integration
- RGB-based interaction
- Wekinator machine learning
- Processing communication
- Custom Arduino behaviours
- Sound
- Character-driven performances

Credit for the original BABOT platform remains with its original creator and contributors.

Authors

Sealia is a student project created for the module Cyber Physical Systems at Imperial College. London, UK. 2026.